

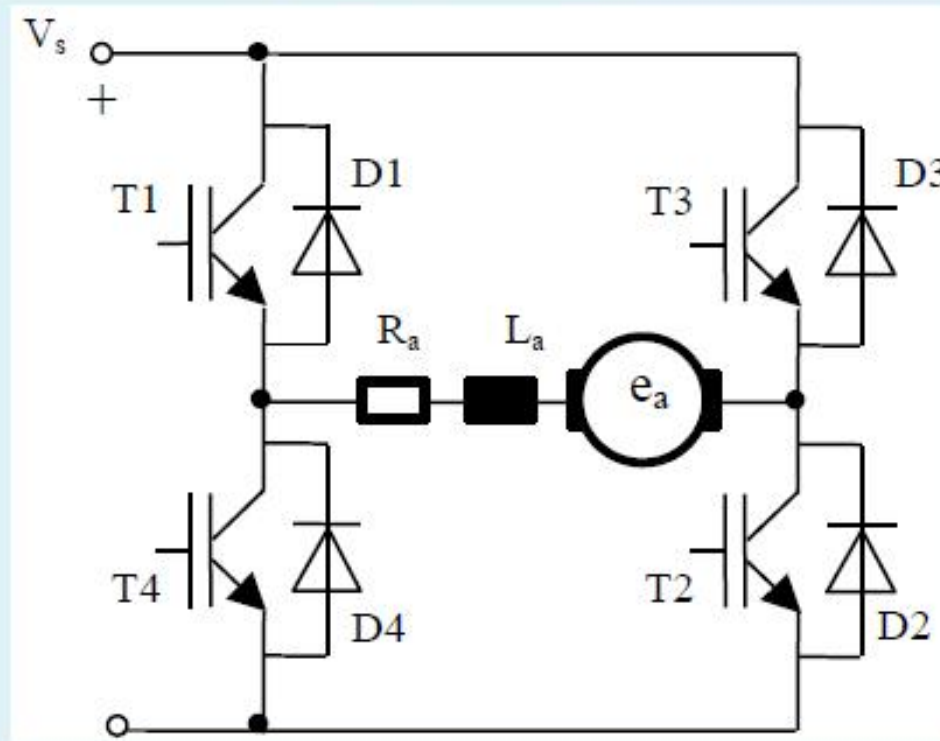
Question 1

Not yet
answered

Marked out of
1.00

Flag question

A DC machine is supplied from a DC supply via a four-quadrant DC chopper (H-Bridge) as shown in the figure below. How is the reverse motoring speed control in the third quadrant achieved?



Select one:

- ☐ By continuously turning on T3 and T4
- ☐ By turning on T3 continuously and turning on and off T4 and D1 interchangeably
- ☐ By continuously turning on D3 and D4
- ☐ The answers in 'a' and 'b' are the possible modes

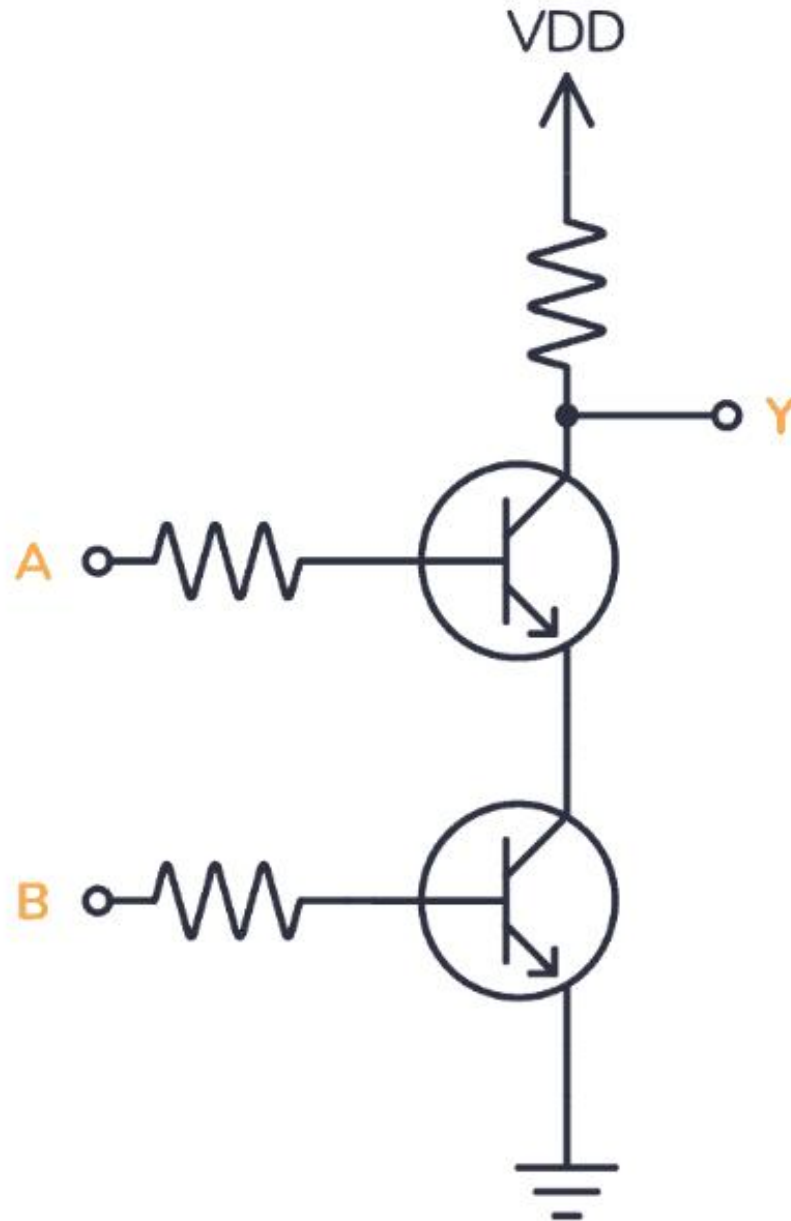
Question 2

Not yet
answered

Marked out of
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What Logic function does the following Transistor Logic Circuit exhibit?



Select one:

☐ NAND

Question **3**

Not yet
answered

Marked out of
1.00

🚩 Flag question

Using the first-order pressure transducer transfer function below, calculate the rise time?

$$G(s) = \frac{3}{(2s + 1)}$$

Select one:

- ☐ 4.9 s
- ☐ 3.5 s
- ☐ 3.2 s
- ☐ 4.4 s

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Question **4**

Not yet
answered

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A voltage $v(t) = 6e^{-2t}$ is applied at $t=0$ to a series RL circuit with $L=1\text{H}$. If $i(t) = 6e^{-2t} - 6e^{-3t}$ then R will have a value of

Select one:

- ☐ 2/3 ohm
- ☐ 3 ohm
- ☐ 2 ohm
- ☐ 0.6 ohm

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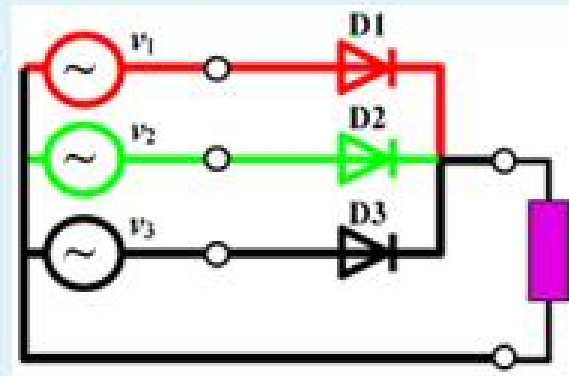
Question 5

Not yet
answered

Marked out of
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In the circuit shown in the figure below, the balanced three-phase sinusoidal voltages v_1 , v_2 , and v_3 have an rms values of 220V at 50Hz. Assume current in the dc side with RL load to be continuous and ripple-free with a value of 12.86A. What is the average value of the output voltage, V_{dc} ?



Select one:

- ☐ 257.3 V
- ☐ 181.9 V
- ☐ 514.6 V
- ☐ 148.6 V

Question **6**

Not yet
answered

Marked out of
1.00

🚩 Flag question

What will be the higher cutoff frequency of a two-stage amplifier if the higher cutoff frequency of each stage is 4MHz?

Select one:

- ☐ 2.57 MHz
- ☐ 6.22 MHz
- ☐ 2 MHz
- ☐ 8 MHz

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Question **7**

Answer saved

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1.00

🚩 Flag question

What are the conditions to be fulfilled for a thyristor (SCR) to be turned on?

Select one:

- ☐ The anode potential should be positive with respect to the cathode potential and a positive gate current pulse should be supplied until the thyristor current is above the latching current.
- ☒ The cathode potential should be positive with respect to the anode potential and a positive gate current pulse should be supplied until the thyristor current is above the holding current.
- ☐ The thyristor should be forward biased
- ☐ The thyristor should be reverse biased

[Clear my choice](#)

Question 8

Not yet
answered

Marked out of
1.00

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A Si sample ($n_i = 1 \times 10^{10}/\text{cm}^3$) at 300 K is uniformly doped with

$N_D = 10^{16} \text{ cm}^{-3}$ donors. if the sample is later doped with $N_A = 5 \times 10^{16} \text{ cm}^{-3}$, the concentration of electrons, holes and the type of material, respectively, are:

Select one:

- ☐ $n \cong 10^{16} \text{ cm}^{-3}$, $p \cong 5 \times 10^{16} \text{ cm}^{-3}$, p-type
- ☐ $n \cong 5 \times 10^{16} \text{ cm}^{-3}$, $p \cong 10^{16} \text{ cm}^{-3}$, n-type
- ☐ $n \cong 25 \times 10^2 \text{ cm}^{-3}$, $p \cong 4 \times 10^{16} \text{ cm}^{-3}$, p-type
- ☐ $n \cong 4 \times 10^{16} \text{ cm}^{-3}$, $p \cong 25 \times 10^2 \text{ cm}^{-3}$, n-type

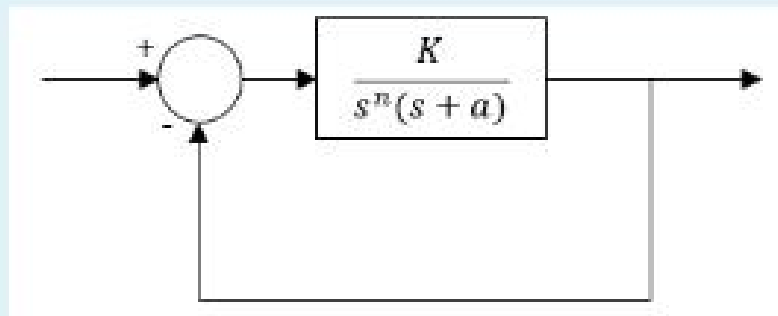
Question 9

Answer saved

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In the figure below, for what value of n will the system have zero steady state step input error?



Select one:

- ☐ 0
- ☒ 1
- ☐ -1
- ☐ -2

[Clear my choice](#)

Question **10**

Answer saved

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1.00

🚩 Flag question

Which of the following methods is easily applicable to control the speed of the squirrel-cage induction motor

Select one:

- ☐ By changing the number of stator poles
- ☐ Rotor rheostat control
- ☐ By operating to motors in cascaded mode
- ☒ By Injecting e.m.f in the rotor circuit

[Clear my choice](#)

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Question **11**

Not yet
answered

Marked out of
1.00

🚩 Flag question

If the driving point admittance function of a one-port network is $Y(s) = \frac{ks}{s+\alpha}$, it can be realized using

Select one:

- ☐ Series combination of R, L
- ☐ Parallel combination of R and L
- ☐ Series combination of R and C
- ☐ Parallel combination of R and C

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Question **12**

Not yet
answered

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Which of the following is an example of an active network component?

Select one:

- ☐ Resistor
- ☐ Capacitor
- ☐ Diode
- ☐ Operational Amplifier

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Question **13**

Not yet
answered

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Consider a 16-bit single Precision Floating number representation with the below bit allocation. Which of the following numbers is the smallest?



Select one:

- ☐ 0 10000001 1000000
- ☐ 1 10000001 1000000
- ☐ 1 11111111 1000000
- ☐ 1 01111111 1000000

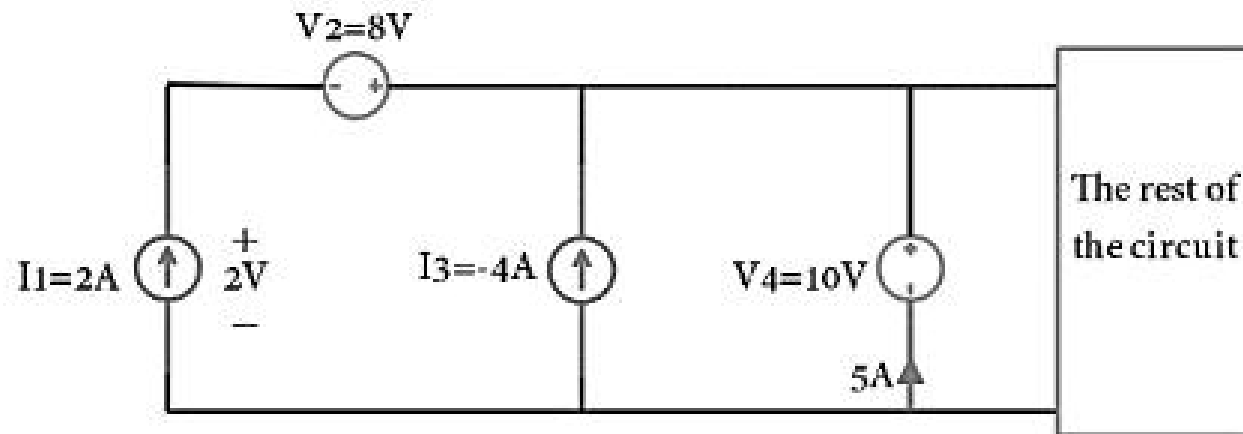
Question **14**

Answer saved

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In the figure below, which source is absorbing power?



Select one:

- ☐ I_1
- ☒ V_2
- ☐ I_3
- ☐ V_4

[Clear my choice](#)

Question **15**

Not yet
answered

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Which one of the following is not the characteristic of an ideal switch?

Select one:

- ☐ Zero voltage drop during on state
- ☐ Finite non zero time during transition between the two states
- ☐ Limited voltage rating during blocking or off state
- ☐ 'b' and 'c'

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Question **16**

Not yet
answered

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With negative feedback in a closed loop control system, the system sensitivity to parameter variation:

Select one:

- ☐ Becomes infinite
- ☐ Becomes zero.
- ☐ Decreases
- ☐ Increases

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Question **17**

Not yet
answered

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Which of the following statement is false concerning the root locus on the real axis for a closed-loop system with a loop gain

$$G(s)H(s) = \frac{k(s + 10)}{(s + 2)(s + 5)}$$

Select one:

- ☐ The root locus starts from $s = -2$, $s = -10$ and $s = -5$
- ☐ It has breakaway point between $s = -2$ and $s = -5$
- ☐ It has break-in point between $s = -10$ and $s = -20$
- ☐ Interval on the real axis between $s = -2$ and $s = -5$ is part of the root locus.

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Question **18**

Not yet
answered

Marked out of
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🚩 Flag question

Which of the following numbers differs from others in terms of value.

Select one:

- ☐ 0b110011110100010
- ☐ 0x67A2
- ☐ 26531 to the base 10
- ☐ 63642 to the base 8

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Question **19**

Answer saved

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Which one of the following is not the main purpose of biasing an amplifier?

Select one:

- ☐ To establish a fixed level of current and voltage
- ☐ To make the amplifier stable
- ☐ To amplify the DC supply
- ☒ To establish the operating point of the amplifier

[Clear my choice](#)

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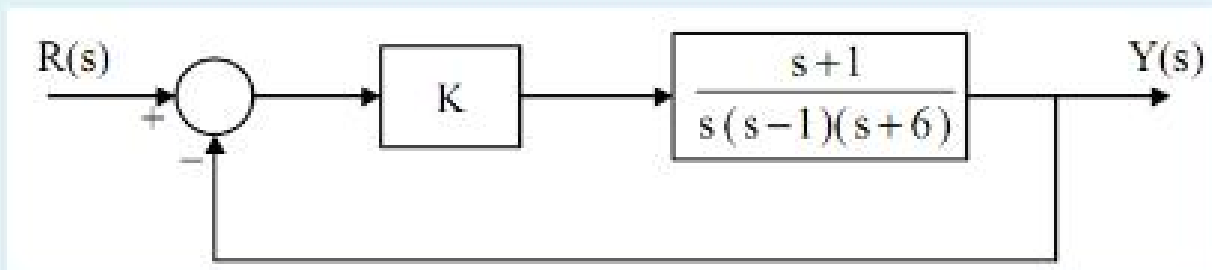
Question **20**

Not yet
answered

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For the closed system shown in the figure below, the range of the values of K for stability is



Select one:

- ☐ $0 < K < 7.5$
- ☐ $K > 7.5$
- ☐ $K > 15$
- ☐ $K > 0$

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Question **21**

Not yet
answered

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🚩 Flag question

In voltage divider bias, operating point is 3 V, 2 mA. If $V_{CC} = 9\text{ V}$, $R_C = 2.2\text{ kohm}$, what is the value of R_E ?

Select one:

- ☐ 2000 ohm
- ☐ 1400 ohm
- ☐ 800 ohm
- ☐ 1600 ohm

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Question **22**

Answer saved

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A unit step current is applied through a 1-H inductor. The voltage across the inductor is:

Select one:

- ☐ $u(t)$
- ☐ $\text{sgn}(t)$
- ☐ $e^{-t}u(t)$
- ☒ $\delta(t)$

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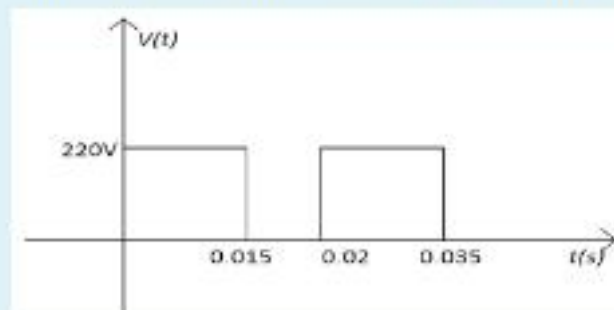
Question **23**

Not yet
answered

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Consider the square wave voltage source in the figure below. If you are to replace this source with a DC source of equal power for a resistive source, what will be the value of the DC voltage?



Select one:

- ☐ 190.5V
- ☐ 110V
- ☐ 440V
- ☐ 210.4V

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Question **24**

Not yet
answered

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A two port network is described by the relations:

$$V_1 = 2V_2 + 0.5I_2$$

$$I_1 = 2V_2 + I_2$$

what is the value of the h_{22} parameter of the network?

Select one:

- ☐ 1 ohm
- ☐ 2 ohm
- ☐ -2 ohm
- ☐ 4 ohm

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Question **25**

Not yet
answered

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🚩 Flag question

Karnaugh Maps are graphical methods used to reduce/ simplify logic functions. For the following 4 variable K-Map, what is the resulting logic function in its most reduced form?

		cd				
		00	01	11	10	
ab	00	1		0	1	1
	01	1		1	0	1
	11	1		1	0	1
	10	1		0	1	1

Select one:

- ☐ $ab+ab'+b$
- ☐ $d'+b'c+bc'$
- ☐ $c'd'+cd'+bc'd+a'b'c+ab'c$
- ☐ 1

Question **26**

Answer saved

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Laplace transform of the output response of a linear system is the system transfer function when the input is

Select one:

- ☐ Unit step signal
- ☐ a ramp signal
- ☒ an Impulse signal
- ☐ a sinusoidal signal

[Clear my choice](#)

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Question **27**

Answer saved

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If the field of a dc shunt motor gets opened while the motor is running

Select one:

- ☐ The speed of the motor will reduce
- ☐ The armature current will reduce
- ☐ The motor continue to rotate at constant speed
- ☒ The motor will attain dangerously high speed

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Question **28**

Answer saved

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What is the vital component of a transformer that gives protection to a transformer from different internal faults like Short circuit and inter-turn?

Select one:

- ☐ Buchholz relay
- ☒ HRC fuse
- ☐ Silica gel breather
- ☐ Thermal relay

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Question **29**

Not yet
answered

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$y(t) = x(t/5)$ is

Select one:

- ☐ Amplitude scaled signal by factor of 1/5
- ☐ Time-shifted signal
- ☐ Expanded signal
- ☐ Compressed signal

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Question **30**

Answer saved

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1.00

🚩 Flag question

How does the location of poles and zeros in a pole-zero diagram affect the stability of a system?

Select one:

- ☐ Systems with poles in the right half-plane are stable
- ☒ Systems with poles in the left half-plane are stable
- ☐ Systems with multiple poles on the $j\omega$ axis are stable
- ☐ The location of poles and zeros has no effect on system stability

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Question **31**

Not yet
answered

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🚩 Flag question

Which one of the following is correct about the full-wave bridge rectifier?
Assume Si diodes with voltage drop V_D and the amplitude of the input voltage waveform is V_m .

Select one:

- ☐ The amplitude of the output waveform is $V_m - V_D$
- ☐ The PIV rating of each silicon diode is V_m
- ☐ The PIV rating of all the diodes are the same.
- ☐ The DC component of the output waveform is $0.318(V_m - V_D)$

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Question **32**

Not yet
answered

Marked out of
1.00

🚩 Flag question

If the characteristic equation of the closed loop system is $s^2 + 2s + 2 = 0$, then the system is:

Select one:

- ☐ Over damped
- ☐ Critically damped
- ☐ Undamped
- ☐ Underdamped

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Question **33**

Not yet
answered

Marked out of
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🚩 Flag question

How would you represent -1 with 8-bits 2's complement Representation?

Select one:

- ☐ 10000001
- ☐ 11111110
- ☐ 11111111
- ☐ 00000001

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Question **34**

Not yet
answered

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The centroid in the root locus is a point where

Select one:

- ☐ The branches of the root locus intersect with the imaginary axis.
- ☐ The branches of the root locus tend to infinity.
- ☐ The asymptotes cross the real axis.
- ☐ The branches of the root locus terminate on the real axis

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Question **35**

Answer saved

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A Shunt generator running at 1000r.p.m has generated e.m.f as 200V.If the speed increases to 1200r.p.m , the generated e.m.f will be

Select one:

- ☐ 150V
- ☐ 175V
- ☒ 240V
- ☐ 290V

[Clear my choice](#)

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Question **36**

Answer saved

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🚩 Flag question

A packet of resistors with a nominal value of 400 ohm and a manufacturer tolerance of 5% can be purchased from an electrical component store. If one resistance is randomly selected, What is the maximum resistance value that this particular resistor is likely to have?

Select one:

- ☐ 450 ohm
- ☐ 380 ohm
- ☒ 420 ohm
- ☐ 390 ohm

[Clear my choice](#)

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Question **37**

Not yet
answered

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🚩 Flag question

Which one of the following is not correct about class A power amplifier?

Select one:

- ☐ It has minimum distortion
- ☐ Its power conversion efficiency is limited to below 25%
- ☐ It is biased at zero dc current
- ☐ The transistor conducts for the entire cycle of the input signal

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Question **38**

Not yet
answered

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Suppose you want to determine R_{th} (Thevenin) for a circuit by observing the loading effect of the load resistor R_L . When the load is connected, the voltage across the load is 75% of the voltage when the circuit is unloaded. How much is the load resistance compared to the R_{th} ?

Select one:

- ☐ $R_L = 3 R_{th}$
- ☐ $R_L = R_{th}/3$
- ☐ $R_L = 0.3 R_{th}$
- ☐ $R_L = 0.7 R_{th}$

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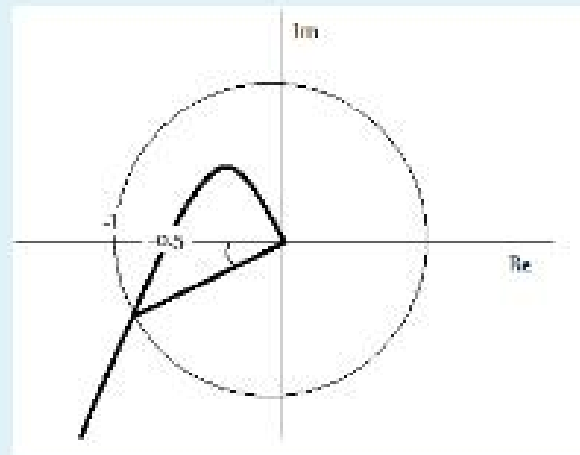
Question **39**

Not yet
answered

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Flag question

In figure below is the polar plot of the loop transfer function of certain unity feedback control system. By referring to the figure, what is the gain margin of the system?



Select one:

- ☐ 0.5
- ☐ -2
- ☐ 60°
- ☐ 2

Question **40**

Answer saved

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Which of the following are odd functions?

Select one:

☐ $\sin t + \cos t$

☐ $t \sin t$

☐ $t \ln t$

☒ $t^3 \cos t$

[Clear my choice](#)

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Question **41**

Not yet
answered

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🚩 Flag question

What is a logic function with two inputs that generates FALSE whenever the two inputs are Similar, and TRUE otherwise?

Select one:

- ☐ NOR
- ☐ XNOR
- ☐ XOR
- ☐ AND

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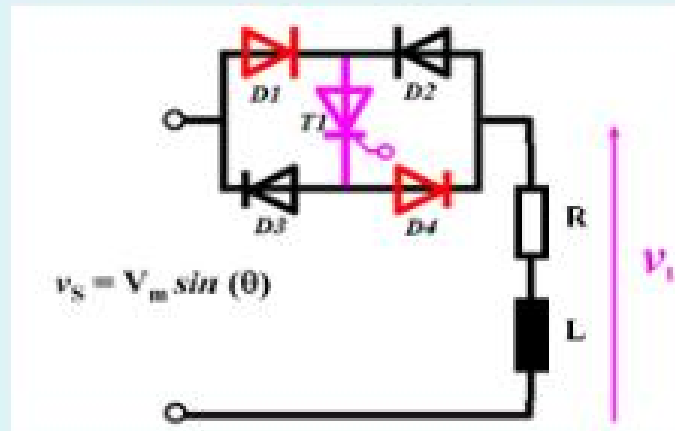
Question 42

Not yet
answered

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1.00

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For the circuit in the figure, the input ac voltage is 220V at 50Hz, $R = 10\Omega$ and $L = 32\text{ mH}$. What is the control range of the firing angle, α ?



Select one:

- ☐ 45.15 degree - 180 degree
- ☐ 0 degree – 180 degree
- ☐ 45.15 degree - 90 degree
- ☐ 30 degree - 180 degree

Question **43**

Not yet
answered

Marked out of
1.00

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A proportional controller leads to:

Select one:

- ☐ Zero steady state error for step input for type 1 system
- ☐ Zero steady state error for step input for type 0 system
- ☐ Infinite steady state error for step input for a type 1 system
- ☐ Finite steady state error for step input for type 1 system

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Question **44**

Answer saved

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The Laplace transform of $u(t - 3)$ is:

Select one:

☐ $1/s + 3$

☒ e^{-3s}/s

☐ $1/s - 3$

☐ e^{3s}/s

[Clear my choice](#)

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Question **45**

Not yet
answered

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🚩 Flag question

The current in an electric circuit rises exponentially as given by

$i(t) = 10(1 - e^{-2t})$ A. How much charge flows through the circuit between $0 \leq t \leq 0.25$ s.

Select one:

- ☐ 0.771C
- ☐ 0.533C
- ☐ 0.630C
- ☐ 1.0C

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Question **46**

Not yet
answered

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🚩 Flag question

Which of the following approaches to measuring low resistance is most common?

Select one:

- ☐ Ducter ohmmeter method
- ☐ Kelvin double bridge method
- ☐ Potentiometer method
- ☐ Ammeter-voltmeter method

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Question **47**

Not yet
answered

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The system having input $x(t)$ and output $y(t)$ is defined as $y(t) = x(t^2)$. This system is:

Select one:

- ☐ causal, linear and time-varying
- ☐ non-causal, linear and time-variant
- ☐ non causal, non-linear and time-invariant
- ☐ causal, non-linear and time-varying

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Question **48**

Not yet
answered

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The integral $\int_{-\infty}^{\infty} \frac{10\delta(\omega - 1)}{4 + \omega^2} d\omega$ gives:

Select one:

- ☐ 0
- ☐ 2
- ☐ 2.5
- ☐ *Infinite*

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Question **49**

Not yet
answered

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Suppose you are given a task to build an 8x256 Decoder only by cascading smaller decoders as building block. Which of the following combinations is a valid combination?

Select one:

- ☐ Eight 4x16 Decoders & One 3x8 Decoder
- ☐ Four 6x64 Decoders & One 2x4 Decoder
- ☐ Sixteen 3x8 Decoders & One 5x32 Decoder
- ☐ Eight 5x32 Decoders & One 4x16 Decoder

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Question **50**

Answer saved

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How does increasing the order of a filter affect the complexity of the circuit?

Select one:

- ☐ It decreases the complexity of the circuit.
- ☐ It has no effect on the complexity of the circuit.
- ☒ It increases the complexity of the circuit.
- ☐ It depends on the type of filter being designed.

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Question **51**

Answer saved

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If a dc generator fails to build up , the probable cause could not be ,

Select one:

- ☐ Imperfect brush contact
- ☐ Field resistance less than the critical resistance
- ☒ No residual magnetism in the generator
- ☐ Faulty shunt connections tending to reduce the residual magnetism

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Question **52**

Not yet
answered

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🚩 Flag question

A step-up chopper is to provide a 110V across a 20 W load from a 24V DC supply (Battery). The chopper inductance is 20 micro henry and the chopper frequency is 200 kHz. Assume that the circuit is in steady state and the circuit operates in continuous current conduction mode. Also consider a capacitor across the load that has a value large enough for ripple free output voltage. What are the values of the maximum and minimum battery currents?

Select one:

- ☐ $I_{\max} = 7.84\text{A}$ and $I_{\min} = 3.15\text{A}$
- ☐ $I_{\max} = 27.55\text{A}$ and $I_{\min} = 22.86\text{A}$
- ☐ $I_{\max} = 34.58\text{A}$ and $I_{\min} = 29.89\text{A}$
- ☐ All the given answers are incorrect

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Question **53**

Not yet
answered

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1.00

🚩 Flag question

Which of the following is a lag compensator?

Select one:

☐ $\frac{s+0.1}{s+1}$

☐ $\frac{5s+1}{s+1}$

☐ $\frac{0.5s+1}{s+1}$

☐ $5s + 1$

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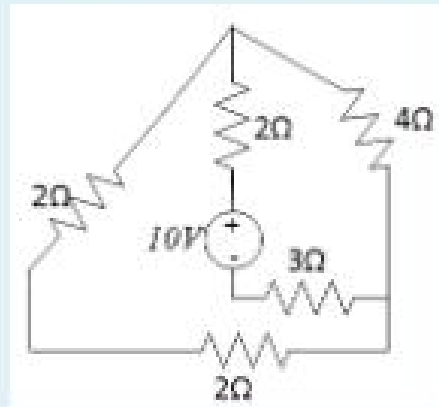
Question **54**

Answer saved

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Flag question

What is the equivalent resistance seen by the source in the following electric circuit in the figure below?



Select one:

- ☐ 2 ohm
- ☒ 7 ohm
- ☐ 10 ohm
- ☐ 8 ohm

[Clear my choice](#)

Question **55**

Not yet
answered

Marked out of
1.00

🚩 Flag question

If two two-port networks are connected in series, and if the port current requirement is satisfied, which of the following is true?

Select one:

- ☐ The z-parameter matrices add.
- ☐ The y-parameter matrices add.
- ☐ The ABCD-parameter matrices add.
- ☐ None of these

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Question **56**

Not yet
answered

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1.00

🚩 Flag question

In a Linear Variable Differential Transducer(LVDT), the two secondary voltages;

Select one:

- ☐ Are indepemndent of the core position
- ☐ Vary equally depending on core position
- ☐ Vary unequally depending on core position
- ☐ Are always in phase quadrature

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Question **57**

Answer saved

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 Flag question

What is the magnetic field's force on Hall Effect transducers?

Select one:

- ☒ Lorentz force
- ☐ Hall Effect force
- ☐ Magnetic force
- ☐ Electric force

[Clear my choice](#)

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Question **58**

Answer saved

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 Flag question

The purpose of capacitors in a transistor amplifier is to

Select one:

- ☐ Protect the transistor
- ☐ Cool the transistor
- ☒ Couple or bypass a.c. component
- ☐ Provide biasing

[Clear my choice](#)

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Question **59**

Not yet
answered

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1.00

🚩 Flag question

Which of the following accurately sums up the inductive transducer's disadvantage in terms of measuring displacement?

Select one:

- ☐ Act of electromagnetic force of attraction
- ☐ Lower sensitivity
- ☐ Requirement of large displacement
- ☐ None of the mentioned

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Question **60**

Answer saved

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A micro industry owner who uses single phase 220V 50 Hz power supply consumes power of 5kW at pf of 0.7. The owner wants to reduce his energy bill by installing power factor corrector. What is the capacitance value of the capacitor bank to make the pf 0.95?

Select one:

- ☐ 227mF
- ☐ 100mF
- ☐ 307mF
- ☒ 1mF

[Clear my choice](#)

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Question **61**

Answer saved

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🚩 Flag question

The current measured through a circuit element connected across a voltage source of $v(t) = 10\cos(314t + 20^\circ)V$ is $i(t) = 2\cos(314t + 70^\circ)A$. What is the nature of the circuit element?

Select one:

- ☐ Inductive
- ☒ Capacitive
- ☐ Resistive
- ☐ undetermined

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Question **62**

Not yet
answered

Marked out of
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🚩 Flag question

Consider the CMOS circuit with Transistors T1, T2, T3 and T4 illustrated below. For the logic inputs $AB = 10$, which of the following statements is True?

Select one:

- ☐ T1 will be turned ON & T2 will be turned OFF.
- ☐ T2 and T3 will be turned ON & while T1 and T4 are turned OFF.
- ☐ T1 and T2 will never be turned ON simultaneously for any input combination.
- ☐ The output Y will be LOW

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Question **63**

Answer saved

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🚩 Flag question

Given a function $f(t)=A[u(t)-u(t-T)]$ where A is a constant, $u(t)$ is a unit step function and T is a time delay. Then the Laplace transform $F(s)$ of $f(t)$ is given by:

Select one:

- ☒ $A\left(\frac{1 - e^{-Ts}}{s}\right)$
- ☐ $A(1 - e^{-Ts})$
- ☐ $A\left(\frac{1}{s} - e^{-Ts}\right)$
- ☐ Ae^{-Ts}

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Question **64**

Not yet
answered

Marked out of
1.00

🚩 Flag question

Which of the following signal is neither an energy nor a power signal?

Select one:

- ☐ Unit step signal
- ☐ Impulse signal
- ☐ Ramp Signal
- ☐ Exponential signal

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Question **65**

Not yet
answered

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🚩 Flag question

Which of the following is correct for moisture transducers?

Select one:

- ☐ Dielectric constant of pure water greater than other materials
- ☐ Dielectric constant of pure water much less than other materials
- ☐ Dielectric constant of pure water and of other materials are equal
- ☐ None of the mentioned

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Question **66**

Not yet
answered

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🚩 Flag question

What is the purpose of feedback in an amplifier circuit?

Select one:

- ☐ To increase the overall gain of the amplifier
- ☐ To reduce distortion in the output signal
- ☐ To stabilize the amplifier's performance and reduce sensitivity to component variations
- ☐ To prevent the amplifier from generating noise

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Question **67**

Answer saved

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The armature reaction effect in a synchronous machine depends on

Select one:

- ☐ Load current
- ☐ Power factor of the load
- ☐ Speed of the machine
- ☒ a and b

[Clear my choice](#)

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Question **68**

Answer saved

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 Flag question

It is known that most industrial motors are squirrel cage induction motors. The rotors of these motors are slightly skewed in order to

Select one:

- ☐ Reduce windage loss
- ☒ reduce eddy current
- ☐ reduce accumulation of dirt and dust
- ☐ reduce magnetic hum

[Clear my choice](#)

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Question **69**

Answer saved

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🚩 Flag question

The impulse response of a continuous time system is given by $h(t) = \delta(t-2) + \delta(t-4)$. The value of the step response at $t=3$ is

Select one:

- ☐ 1
- ☐ 0
- ☒ 2
- ☐ 3

[Clear my choice](#)

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Question **70**

Not yet
answered

Marked out of
1.00

🚩 Flag question

An accelerometer, a second-order transducer has a natural angular frequency of 2.0 rad/s and a damped frequency of 1.8 rad/s . What are its damping factor and percentage overshoot, respectively?

Select one:

- ☐ 0.52 and 25%
- ☐ 0.71 and 45%
- ☐ 0.44 and 21%
- ☐ 0.27 and 63%

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Question **71**

Answer saved

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🚩 Flag question

Two resistors R_1 and R_2 are made up of a copper wire. The length and area of the first resistor are l and A respectively and that of the second resistor are $0.25l$ and $0.5A$. What is the ratio of the currents (I_{R2}/I_{R1}) if the two resistors are connected across the same voltage sources?

Select one:

- ☒ 0.5
- ☐ 5.0
- ☐ 2.0
- ☐ 0.25

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Question **72**

Answer saved

Marked out of
1.00

🚩 Flag question

Which switching device is with the highest possible switching speed?

Select one:

- ☐ Thyristor (SCR)
- ☒ IGBT
- ☐ Power BJT
- ☐ Power MOSFET

[Clear my choice](#)

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Question **73**

Answer saved

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 Flag question

Which DC motor should not be started on no-load

Select one:

- ☒ Series motor
- ☐ Shunt motor
- ☐ Cumulatively compounded motor
- ☐ Differentially compounded motor

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Question **74**

Not yet
answered

Marked out of
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🚩 Flag question

The efficiency of a transformer will be maximum when

Select one:

- ☐ Copper losses = hysteresis loss
- ☐ Hysteresis loss = eddy current loss
- ☐ Eddy current loss = copper loss
- ☐ Copper loss = iron loss

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Question **75**

Not yet
answered

Marked out of
1.00

🚩 Flag question

The phase difference between the output and input voltages of a CE amplifier is

Select one:

- ☐ 180 degree
- ☐ 0 degree
- ☐ 90 degree
- ☐ 270 degree

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Question **76**

Answer saved

Marked out of
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🚩 Flag question

What is the characteristic feature of a Chebyshev filter?

Select one:

- ☐ A maximally flat passband response
- ☐ An equiripple passband response
- ☒ A linear phase response
- ☐ A constant group delay

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Question **77**

Not yet
answered

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1.00

🚩 Flag question

Consider the plant transfer function $\frac{1}{s^2}$. What is the proportional controller for the system to have gain cross over frequency of 2rad/sec.

Select one:

- ☐ 8
- ☐ 3
- ☐ 4
- ☐ 2

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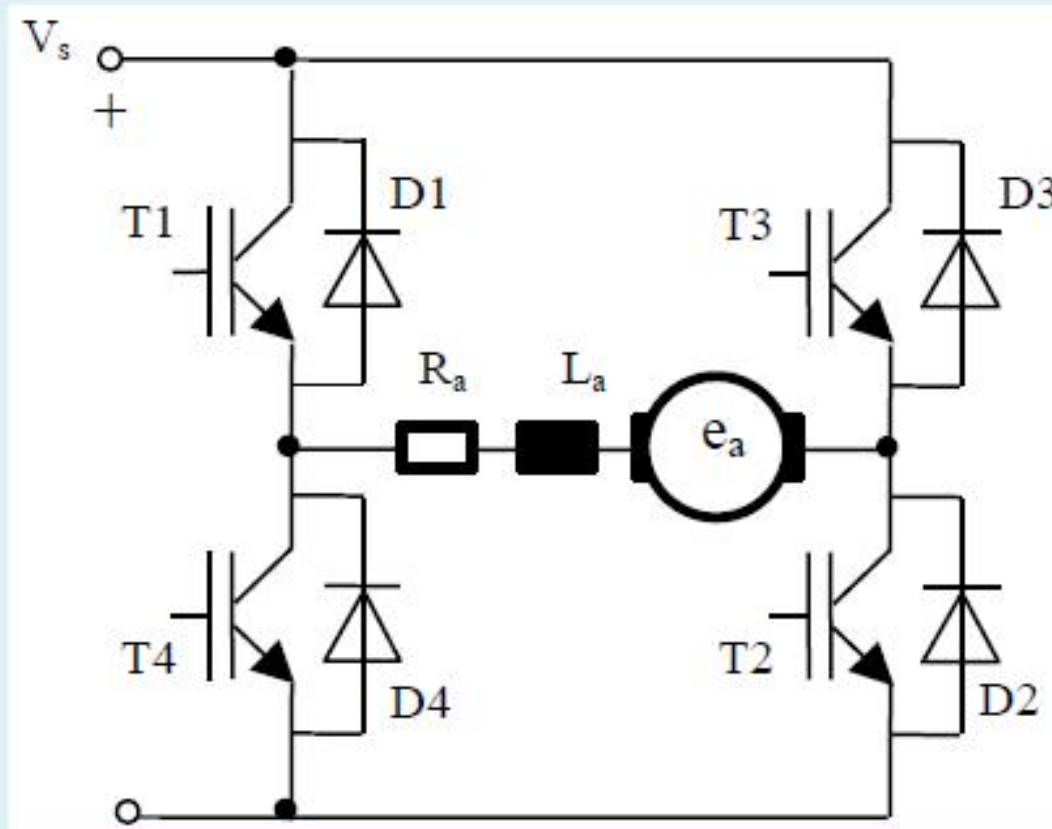
Question **78**

Not yet
answered

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Flag question

A DC machine is supplied from a DC supply via a four-quadrant DC chopper (H-Bridge) as shown in the figure below. How is the forward motoring speed control in the first quadrant achieved?



Select one:

- ☐ By turning on T1 continuously and turning on and off T2 and D3 interchangeably
- ☐ By turning on T2 continuously and turning on and off T1 and D4 interchangeably

Question **79**

Not yet
answered

Marked out of
1.00

🚩 Flag question

In a six pulse (three phase full bridge) rectifier, the lowest order harmonic component of the source current which is available next to the fundamental component is_____.

Select one:

- ☐ The second order harmonic
- ☐ The third order harmonic
- ☐ The fifth order harmonic
- ☐ The seventh order harmonic

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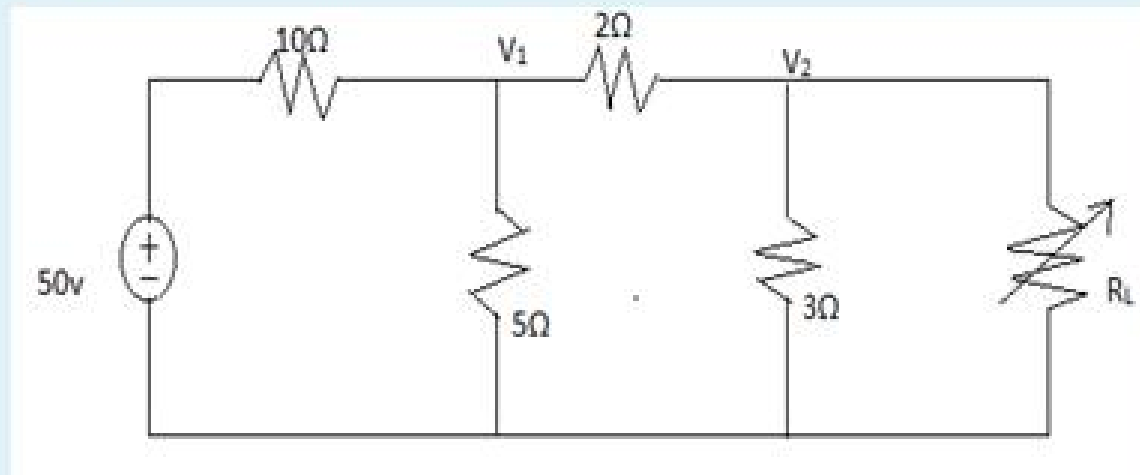
Question **80**

Answer saved

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🚩 Flag question

Determine the maximum power delivered to the load in the network given below



Select one:

- ☐ 4.68W
- ☒ 5.75W
- ☐ 3.16W
- ☐ 6.84W

[Clear my choice](#)

Question **81**

Answer saved

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🚩 Flag question

Among the following statements, regarding Common functional Logic blocks, Identify the one that is *Invalid*.

Select one:

- ☐ Multiplexers have multiple input lines and one output line.
- ☐ In an $n \times m$ binary decoder, the number of input and output lines are related as $m = 2$ raised to n .
- ☒ Demultiplexers can be used to connect multiple input lines into a single shared channel/line.
- ☐ An Encoder can be used to collapse/ reduce the number of bits required to represent a multi-bit digital data.

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Question **82**

Answer saved

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🚩 Flag question

What will happen if a DC supply is given to a transformer?

Select one:

- ☐ It will provide constant DC at the output
- ☐ Its primary winding will burn
- ☐ Its secondary winding will burn
- ☒ Nothing will happen to the transformer

[Clear my choice](#)

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Question **83**

Answer saved

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🚩 Flag question

Which of the following responses of an LTI system does not depend on initial conditions?

Select one:

- ☐ Natural response
- ☐ free response
- ☒ forced response
- ☐ total response

[Clear my choice](#)

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Question **84**

Not yet
answered

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1.00

🚩 Flag question

A voltmeter's 0-15 V range, which records a value of 83 V, has a 1% full-scale accuracy guarantee. Determine the limiting error as a percentage.

Select one:

- ☐ 0.55 %
- ☐ 5.53 %
- ☐ 0.18 %
- ☐ 1.81 %

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Question **85**

Not yet
answered

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🚩 Flag question

In which region should an N channel Field Effect Transistor be operated to amplify an input signal?

Select one:

- ☐ Triode (Ohmic) Region
- ☐ Saturation Region
- ☐ Cutoff Region
- ☐ None of the above

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Question **86**

Not yet
answered

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🚩 Flag question

In the most straight forward implementation of an n-bit Parallel-In Parallel-Out Register, which flip-flops should be used?

Select one:

- ☐ Toggle Flip Flops (T)
- ☐ Delay Flip Flops (D)
- ☐ JK Flip Flops (JK)
- ☐ Q Flip Flops (Q)

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Question **87**

Not yet
answered

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🚩 Flag question

What is the maximum voltage or peak inverse voltage (PIV) blocked by the diodes in three phase half wave diode rectifiers?

Select one:

- ☐ The maximum value of the supply line to line voltage
- ☐ The maximum value of the supply line to neutral voltage
- ☐ The root mean square (rms) value of the supply line to line voltage
- ☐ The square root of three times the root mean square (rms) value of the supply line to neutral voltage

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Question **88**

Not yet
answered

Marked out of
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🚩 Flag question

Which of the following is correct for toroid-type coils transducers?

Select one:

- ☐ Toroid can be used for the measurement of displacement
- ☐ Toroid cannot be used for measuring displacement
- ☐ Toroid can be used for measuring both displacement and velocity
- ☐ None of the mentioned

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Question **89**

Answer saved

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In a transistor, $I_C = 100 \text{ mA}$ and $I_E = 100.2 \text{ mA}$. The value of beta is

Select one:

- ☐ 100
- ☐ 50
- ☒ about 1
- ☐ 200

[Clear my choice](#)

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Finish attempt ...